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Nomenclature

Physical constants

ε_0	Permittivity of free space	
μ_0	Permeability of free space	
a_0	Bohr radius	
c	Speed of light in vaccuum	
e	Elementary charge	
h	Planck constant	
k_B	Boltzmann constant	
m_e	Electron mass	
N_A	Avogadro's number	
$\mathcal{R}$	Rydberg energy	

Scalars

α	Attenuation coefficient	$\mathrm{m^{-1}}$
Γ	Reaction rate density	$\mathrm{m^{-3}\,s^{-1}}$
γ	Adiabatic index	1
ε_r	Relative permittivity	1
η	Refractive index	1
μ	Dynamic viscosity	$\mathrm{Pa\,s}$
μ_r	Relative permeability	1
ν	Frequency	Hz
ρ	Charge density	$\mathrm{C\,m^{-3}}$
ρ	Mass density	$\mathrm{kg\,m^{-3}}$
σ	Reaction cross section	$\mathrm{m^2}$
Φ	Quantum yield	1
ϕ	Scalar flux	$\mathrm{m^{-2}\,s^{-1}}$
ω	Angular frequency	$\mathrm{rad\,s^{-1}}$
A	Area	$\mathrm{m^2}$
$\mathcal{E}$	Particle internal energy	J
ρE	Internal energy density	$\mathrm{J\,m^{-3}}$
E	Specific internal energy	$\mathrm{J\,kg^{-1}}$
$\mathcal{H}$	Particle enthalpy	J
ρH	Enthalpy density	$\mathrm{J\,m^{-3}}$
H	Specific enthalpy	$\mathrm{J\,kg^{-1}}$
I	Intensity	$\mathrm{W\,m^{-2}}$
k	Thermal conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
M	Molar mass	$\mathrm{kg\,mol^{-1}}$
m	Mass	kg
N	Number	1
n	Number density	$\mathrm{m^{-3}}$
p	Pressure	Pa
q'''	Volumetric heat generation rate	$\mathrm{W\,m^{-3}}$
S	Reaction rate coefficient	varies
s	Distance	m
T	Temperature	K
t	Time	s
$\mathcal{U}$	Particle potential energy	J
V	Volume	$\mathrm{m^3}$
v	Speed	$\mathrm{m\,s^{-1}}$

Vectors

$\hat{\boldsymbol{\omega}}$	Direction	1
$\mathbf{a}$	Acceleration	$\mathrm{m\,s^{-2}}$

$\mathbf{B}$	Magnetic field	T
$\mathbf{E}$	Electric field	$\mathrm{V\,m^{-1}}$
$\mathbf{f}$	Force density	$\mathrm{N\,m^{-3}}$
$\mathbf{g}$	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{j}$	Current density	$\mathrm{A\,m^{-2}}$
$\hat{\mathbf{n}}$	Normal direction	1
$\mathbf{q}''$	Heat flux density	$\mathrm{W\,m^{-2}}$
$\mathbf{r}$	Position	m
$\mathbf{S}$	Poynting vector	$\mathrm{W\,m^{-2}}$
$\mathbf{u}$	Velocity	$\mathrm{m\,s^{-1}}$
$\rho\mathbf{u}$	Momentum density	$\mathrm{N\,s\,m^{-3}}$

Matrices

$\mathbf{T}$	Deviatoric stress tensor	Pa

Subscripts

ci	Collisional ionization
ib	Inverse bremsstrahlung
pi	Photoionization
rec	Recombination
γ	Photon
b	Bound electron
e	Free electron
i	Ion

Superscripts

K	Kinetic energy
M	Momentum

Chapter 1

Introduction

Chapter 2

Physics of laser-droplet interactions

Modeling even a single droplet that interacts with a laser beam is more complicated than a typical thermal-hydraulic simulation. Even as the laser cross section is pretty small compared to the droplet surface area, the migration of free charged particles necesitates treating the entire droplet as a plasma. Modeling of plasmas, especially weakly-ionized ones, typically enforces strong coupling between the conservation equations and Maxwell's equations.

Chemically speaking, the laser can depart enough energies to not only remove bound electrons from their orbits but also break molecular bonds and alter the chemical composition of the fluid. In fact, free radicals and light molecular ions are frequently encountered in plasmas if the laser is not strong enough to completely disintegrate molecules into monoatomic ions. However, for simplicity, all ions are assumed to have resulted from neutral molecules each losing a single electron, and thus they have a charge of +1 and the same molecular weight as the parent neutral particle.

2.1 Fluid description of plasmas

A plasma in itself is a collection of charges, each with its own trajectory. The most accurate plasma modelling approach is to track the position and velocity of all charged particles within the simulation domain; however, the expense associated with such a model would be prohibitive. A fluid model approximates individual particles of like charges as a single collective body, as well as neutral particles, with each species being described by a set of fluid equations. As a result, the electrons, ions, and neutral particles form three interpenetrating fluids. The fluids of charged particles are distinct from the neutral thanks to their interactions with the electromagnetic fields, as described in Section 2.2.

For the rest of this paper, the conservative and primitive variables carry the subscript e for electrons and i for ions. Neutral particles typically do not have a subscript; unless a particular term describes the interaction between a neutral and a charged particle (see Section 2.3), then it has a subscript n. The subscript j is a placeholder for either e, i, sometimes n, or γ for photons (see Section 2.4).

2.1.1 Conservation equations

The neutral fluid is described as a compressible, viscous flow. Its conserved variables are described by the Navier-Stokes equations. Without external sources, these equations read

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = 0 \tag{2.1a}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \rho \mathbf{u} \otimes \mathbf{u} = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g} \tag{2.1b}$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot \rho \mathbf{u} H = -\nabla \cdot \mathbf{q}'' + \nabla \cdot [\mathbf{T} \cdot \mathbf{u}] \tag{2.1c}$$

The diffusive terms in the momentum and energy equations are $\nabla \cdot \mathbf{T}$ and $\nabla \cdot \mathbf{q}''$, respectively. The deviatoric stress tensor $\mathbf{T}$ is

$$\mathbf{T} = \mu \left[\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right], \tag{2.2}$$

and the heat flux vector $\mathbf{q}''$ is

$$\mathbf{q}'' = -k \nabla T. \tag{2.3}$$

Properties related to charged particles tend to be formulated in terms of their number density, so it will served the conserved variable instead of mass density. The conversion between the two densities are

$$\rho_j = \frac{M_j}{N_A} n_j. \tag{2.4}$$

Similarly, a different set of conservation equations can be written for charged particles. It is more common to describe a quantity of charges using number density as opposed to mass density, and therefore the equations read:

$$\frac{\partial n_j}{\partial t} + \nabla \cdot n_j \mathbf{u}_j = 0 \tag{2.5a}$$

$$\frac{\partial n_j \mathbf{u}_j}{\partial t} + \nabla \cdot n_j \mathbf{u}_j \otimes \mathbf{u}_j = \frac{-\nabla p_j + \nabla \cdot \mathbf{T}_j}{m_j} + n_j \mathbf{g} \tag{2.5b}$$

$$\frac{\partial n_j \mathcal{E}_j}{\partial t} + \nabla \cdot n_j \mathbf{u}_j \mathcal{H}_j = -\nabla \cdot \mathbf{q}_j'' + \nabla \cdot [\mathbf{T}_j \cdot \mathbf{u}_j] \tag{2.5c}$$

Here, the average energy and enthalpy per particle, $\mathcal{E}$ and $\mathcal{H}$, are used in order to ensure energy density is in the correct units.

2.1.2 Closure relations

For each set of conservation equations, a closer relation is required to close the system. It usually takes form of an equation of state and relates a conserved variable to a set of primitive variables. For neutral particle, the required equations are that of density and energy as functions of pressure and temperature.

$$\rho = \rho(p, T) \tag{2.6a}$$

$$\rho E = \rho(p, T) H(p, T) - p \tag{2.6b}$$

For charged particles following a Maxwellian distribution of velocities, the average energy per particle is related to its temperature by the simple relation

$$\mathcal{E}_j = \frac{3}{2} k_B T_j. \tag{2.7}$$

Pressure is related to number density by

$$\frac{\nabla p_j}{p_j} = \gamma \frac{\nabla n_j}{n_j}, \tag{2.8a}$$

or

$$p_j = C_j n_j^{\gamma}, \tag{2.8b}$$

where the constant C_j is given by a thermodynamic equation of state.

2.2 Electromagnetic effects

2.2.1 Maxwell's equations

Hydrocarbon materials such that one used in this project can be reasonably assumed to be isotropic and homogeneous. This simplifies the Maxwell's equations in matter because it allows two vectors that are difficult to compute, namely the polarization vector $\mathbf{P}$ and the magnetization vector $\mathbf{M}$, to be parallel to the electric and magnetic fields. Moreover, $\mu_r \approx 1$ is assumed since hydrocarbon are non-magnetic materials. This leads to the following form of Maxwell's equations in matter:

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon_r \epsilon_0} \tag{2.9a}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2.9b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.9c}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}_f + \mu_0 \epsilon_r \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2.9d}$$

In plasma, the free charge density ρ_f and current density $\mathbf{j}_f$ can be easily computed using electron and ion information:

$$\rho_f \equiv e(n_i - n_e) \tag{2.10a}$$

$$\mathbf{j}_f \equiv e(n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.10b}$$

Maxwell's Equations (Equation 2.9) are overdetermined; there are 8 scalar equations that only describes 6 scalar variables (the components of $\mathbf{E}$ and $\mathbf{B}$). However, it can be shown that taking the divergence of Equations 2.9c and 2.9d will lead to Equations 2.9b and 2.9a, respectively. Therefore, only Equations 2.9c and 2.9d are necessary to model the $\mathbf{E}$ and $\mathbf{B}$ fields.

2.2.2 Lorentz force

The Lorentz force acts upon a distribution of charged particles has the form

$$\mathbf{f}_j = \pm e n_j \left(\mathbf{E} + \mathbf{u}_j \times \mathbf{B} \right). \tag{2.11}$$

Since electrons are extremely lightweight, its acceleration due to the Lorentz force easily dominates any diffusive or gravitational effects, so they are safe to neglect. The case for ions is a little bit more nuanced; however,

the Lorentz force is still expected to be the main momentum contributor.

Electromagnetic waves themselves carry energy, and therefore capable of doing work on a charged system. According to Equation 2.11, the magnetic field does no work on charged particles as its force component is always perpendicular to the charge velocity. The rate of work done by the electric field is then

$$W_j = \mathbf{f}_j \cdot \mathbf{u}_j = \pm e n_j \mathbf{u}_j \cdot \mathbf{E}, \tag{2.12}$$

and should be equal to the rate of energy gained by the particles. This accounts for the change in kinetic energy of a charged particles under the influence of the electric fields.

2.3 Collisions in plasmas

Collisions are the primary mean that the species in a plasma exchange their mass, momentum, and energy. The collisions described in this section are assumed to be entirely binary; i.e. involving two particles at a time. Primarily there are two interactions that can follow a collision, namely reaction and relaxation. Reactions convert a neutral particles into ions or vice versa, which changes the composition of the plasma. Relaxations are the main process in which a system is driven towards equilibrium in absence of external sources, is achieved by the two particles involved in a collision exchanging their velocities and temperatures at a rate proportional the respective difference, similar to two working fluids in a heat exchanger.

2.3.1 Reactions

When a free electron collides with a bound electron and departs sufficient energy on it, the latter can be removed from its orbit. Known as collisional ionization, this process is responsible for rapid increase of the free electron population. Not all free electrons can cause collisional ionization; it must have a kinetic energy $\mathcal{E}_e$ that exceeds the binding energy $\mathcal{E}_b$ of the bound electron. Under the fluid model of plasma, electrons in any control volume is modeled as having a single particle-averaged energy. In this section, this averaged energy is denoted $\frac{3}{2}k_B T_e$ in this section as opposed to $\mathcal{E}_e$, which instead represents the energy of an individual electron. Collisional ionization still occurs in a volume where $\frac{3}{2}k_B T_e < \mathcal{E}_b$ since there exists a fraction of electron with a sufficient $\mathcal{E}_e$.

Assuming a Maxwellian distribution of velocity, the probability density function (PDF) of electron speed $f(u_e)$ is [3]

$$f(u_e) = \left(\frac{m_e}{2\pi k_B T_e}\right)^{3/2} 4\pi u_e^2 \exp\frac{-\frac{1}{2}m_e u_e^2}{k_B T_e}. \tag{2.13}$$

To convert Equation 2.13 into a PDF in kinetic energy, the following identity is employed

$$f(\mathcal{E}_e)\,|d\mathcal{E}_e| = f(u_e)\,|du_e|\,. \tag{2.14}$$

which requires the derivative

$$\frac{d\mathcal{E}_e}{du_e} = m_e u_e \tag{2.15a}$$

$$\frac{du_e}{d\mathcal{E}_e} = \frac{1}{m_e u_e} = \frac{1}{\sqrt{2 m_e \mathcal{E}_e}} \tag{2.15b}$$

The energy-PDF is therefore

$$f(\mathcal{E}_e) = f(u_e)\left|\frac{du_e}{d\mathcal{E}_e}\right| \tag{2.16a}$$

$$= \frac{2}{\sqrt{\pi}}\sqrt{\frac{\mathcal{E}_e}{(k_B T_e)^3}}\exp\frac{-\mathcal{E}_e}{k_B T_e} \tag{2.16b}$$

The electron-impart ionization cross section is calculated using the Binary-Encounter-Bethe (BEB) model [6]:

$$\sigma_{\mathrm{BEB}} = \frac{4\pi a_0^2 r^2}{t+u+1}\left[\frac{\ln t}{2}\left(1-\frac{1}{t^2}\right)+1-\frac{1}{t}-\frac{\ln t}{t+1}\right], \tag{2.17a}$$

where the following dimensionless energies are used

$$r \equiv \frac{\mathcal{R}}{\mathcal{E}_b} \tag{2.17b}$$

$$t \equiv \frac{\mathcal{E}_e}{\mathcal{E}_b} \tag{2.17c}$$

$$u \equiv \frac{\mathcal{U}}{\mathcal{E}_b} \tag{2.17d}$$

Let Γ_r be the ionization rate density of a reaction r, which has a positive value if ions are created and negative if they are consumed. For collisional ionization, the

rate density is

$$\Gamma_{\text{ci}} = S_{\text{ci}} n_e n \tag{2.18a}$$

$$S_{\text{ci}} \equiv \int_{\mathcal{E}_b}^{\infty} \sigma_{\text{BEB}}(\mathcal{E}_e) v(\mathcal{E}_e) f(\mathcal{E}_e) d\mathcal{E}_e. \tag{2.18b}$$

The rate constant S_{ci} has units of volumetric flow rate (m^3/s).

The reverse process of ionization is recombination, where an ion captures a free electron to become neutral. Recombination, fortunately, does not have an energy threshold, and the reaction rate density is

$$\Gamma_{\text{rec}} = -S_{\text{rec}} n_e n_i. \tag{2.19}$$

Reactions do not just alter the amount of each species in a plasma; it also transfer a portion of the reactant's momentum and energy directly into the product, assuming no enthalpy of reaction. These transfers are typically more important for ions and neutrals than they are for electrons, as the latter have other sources that contribute to their momentum and energy a lot more significantly. Only account for the transfer of conservative variables in reactive processes, the conservation equations read

$$\frac{\partial n_i}{\partial t} = -\frac{\partial n}{\partial t} = \sum_r \Gamma_r \tag{2.20a}$$

$$\frac{\partial n_i \mathbf{u}_i}{\partial t} = -\frac{\partial n \mathbf{u}}{\partial t} = \sum_r \Gamma_r \mathbf{u}_r \tag{2.20b}$$

$$\frac{\partial n_i \mathcal{E}_i}{\partial t} = -\frac{\partial n \mathcal{E}}{\partial t} = \sum_r \Gamma_r \mathcal{E}_r, \tag{2.20c}$$

where the subscript r on a state variable indicates that the variable belongs to the reactant species.

2.3.2 Relaxations

A collision between a charged particle j and a neutral particle results in a force roughly equal to

$$\mathbf{f}_{jn} = \frac{m_j n_j (\mathbf{u}_j - \mathbf{u})}{\tau_{jn}}. \tag{2.21}$$

Here the double subscript jn refers to a transfer from a particle j to a neutral particle. By Newton's third law of motion, $\mathbf{f}_{jn} = -\mathbf{f}_{nj}$. The mean time between collisions τ_{jn} is approximately constant and is the reciprocal of the momentum-transfer collision frequency

$$\nu_{jn}^M \equiv \frac{1}{\tau_{jn}} = n \sigma_{jn}^M \|\mathbf{u}_j - \mathbf{u}\|, \tag{2.22}$$

with σ_{jn}^M denotes the equivalent cross section of momentum transfer between j and n. It is typically a function of charged particle speed; the average value over all velocities in a Maxwellian distribution is typically used in Equation 2.22.

Collisions between two charged particles are Coulomb collisions mediated by the electric field. Like-particle (electron-electron or ion-ion) collisions are often neglected because they preserve the momentum and energy between themselves, and they contribute very little to diffusion. This is in contrast with electron-ion collisions, where there exists a net transfer of momentum (and energy) between the two particles. In a similar formulation as Equation 2.21, the force exerted between an ion and an electron during their collision is

$$\mathbf{f}_{ei} = m_e n_e (\mathbf{u}_i - \mathbf{u}_e) \nu_{ei}^M, \tag{2.23}$$

and the electron-ion momentum-transfer collision frequency ν_{ei}^M is

$$\nu_{ei}^M = C \frac{n_i Z e^4}{(4\pi\varepsilon_0)^2 \sqrt{m_e (k_B T_e)^3}} \ln \Lambda, \tag{2.24}$$

where C is a prefactor on the order of $\mathcal{O}(1)$ that differs depending on the text. Chen gives $C = \pi$, while the FLASH code uses $C = \frac{4}{3}\sqrt{2\pi}$ [1][2]. $C = \pi$ will be used in this model. The Coulomb logarithm $\ln \Lambda$ is a correction factor that accounts for the cumulative effect of many small-angle deflections in a collision and is given by

$$\ln \Lambda \equiv \ln \left[\frac{4\pi (\varepsilon_0 k_B T_e)^{3/2}}{Z e^3 \sqrt{n_e}} \right]. \tag{2.25}$$

In both Equations 2.24 and 2.25, the average ionization number Z can be assumed to be 1.

For all types of collisions, the energy loss due to relaxation term can be expressed as a heat source that flows from particle j to j'

$$q_{jj'}''' = -q_{j'j}''' = \frac{3}{2} n_j \nu_{jj'}^K k_B (T_j - T_{j'}). \tag{2.26}$$

The energy-transfer collision frequency ν^K is related to its momentum counterpart ν^M by

$$\frac{\nu^K_{jj'}}{\nu^M_{jj'}} = \frac{2m_j}{m_j + m_{j'}}. \tag{2.27}$$

The index j in Equation 2.27 refers to the lighter particle out of the pair, so it is always the electron if it is involved in a collision, and the ion in a ion-neutral collision. In the former case, $\nu^K_{ej'} \ll \nu^M_{ej'}$, which means energy is relaxed at a much slower rate than momentum is. Physically, upon colliding with a heavy particle, an electron is frequently scattered before a significant portion of its kinetic energy is lost. In an ion-neutral collision, due to the near-identical mass of the two particles, energy and momentum are relaxed on the same timescale – i.e., $\nu^K_{in} = \nu^M_{in}$.

2.4 Laser interactions

While the model assumes the droplet is at least partially ionized, the plasma state is not formed without a laser source. The laser introduces a fourth particle – the photon – into the system, and it is quickly absorbed by either bound or free electrons. Whereas the term "electrons" has typically referred to free electrons for much of this chapter, in this context it collectively refers to both free and bound electrons.

2.4.1 Energy deposition

This model assumes the laser beam is a line heat source within each droplet even if the laser cross-sectional area is known. The volumetric heat generation, as a result, contains a Dirac delta function $\delta(\boldsymbol{r}_{\text{laser}})$ where $\boldsymbol{r}_{\text{laser}}$ is the laser trajectory. Finite-volume methods blend well with this assumption since the delta function is always integrated out during the formulation.

$$\iiint_{\Gamma \supset \boldsymbol{r}_0} \delta(\boldsymbol{r}_0) dV = 1 \tag{2.28}$$

Therefore, the total energy deposited in a cell that the laser passes through is the difference in laser energy at the cell entrance and exit. If the entrance energy upon cell k is $I_k A$, it will have been attenuated by a factor of $e^{-\alpha_k s_k}$ when exitting the cell – s_k being the distance that the laser beam travels within cell k. Therefore,

$$\iiint_{V_k} q'''_{\text{laser}} dV = I_k A(1 - e^{-\alpha_k s_k}) \tag{2.29}$$

Bound and free electrons differ in how the energy is spent. A free electron, upon absorbing a photon, simply increases in kinetic energy. A bound electron, however, can either be excited or removed from its orbit altogether. If excitation occurs, collisions with the atom causes the electron to rapidly relax and transfer the acquired energy into thermal energy of the bulk liquid; therefore, on the timescale of interest, it is responsible to assume that the energy goes directly into heating the neutral fluid.

With two channels in which laser energy can be deposited, the total attenuation coefficient α is the sum of the bound and free counterparts.

$$\alpha = \alpha_b + \alpha_{\text{ib}}, \tag{2.30}$$

and the fraction of energy that goes into each channel is proportional to the relative magnitude of the two absorption coefficients

$$q_b = I_0 A(1 - e^{-\alpha s}) \frac{\alpha_b}{\alpha} \tag{2.31a}$$

$$q_{\gamma e} = I_0 A(1 - e^{-\alpha s}) \frac{\alpha_{\text{ib}}}{\alpha}. \tag{2.31b}$$

The subscript "ib" in Equation 2.30 refers to inverse bremsstrahlung, a process in which free electrons, in the vincinty of an ion, absorbs energy and starts to accelerate. The double subscript γe infers as if the heat source were due to a "collision" between a photon and a free electron, similar to how other heat sources are denoted in Section 2.3.2. Inverse bremsstrahlung should take over as the dominant absorption process once a sufficiently opaque medium is formed. The attenuation coefficient will increase proportionally to the inverse bremsstrahlung frequency ν_{ib}

$$\alpha_{\text{ib}} = \frac{\nu_{\text{ib}}\eta}{c} = \frac{\omega_e^2}{\omega^2 + \nu_{ei}^2} \frac{\nu_{ei}\eta}{c}, \tag{2.32}$$

where ω_e is the plasma electron frequency and ω is the laser frequency. The former is defined as

$$\omega_e^2 \equiv \frac{n_e e^2}{\varepsilon_0 m_e}. \tag{2.33}$$

When $\omega \gg \nu_{ei}$, ν_{ib} is usually given in the form

$$\nu_{\text{ib}} = \frac{n_e}{n_c}\nu_{ei}, \tag{2.34}$$

and the critical density n_c is defined in terms of ω

$$\omega^2 = \frac{n_c e^2}{\varepsilon_0 m_e}. \tag{2.35}$$

2.4.2 Photoionization

Photoionization can only occur if the photon energy, $h\nu$, exceeds the binding energy $\mathcal{E}_b$. This is simply the photoelectric effect, and it is typically caused by photons in the ultraviolet regime. For a quantum yield $\Phi \in [0, 1]$ that accounts for the number of photoelectrons produced per incident photon, the photoionization rate density is

$$\Gamma_{\text{pi}} = \Phi \frac{q_b'''}{h\nu}, \tag{2.36}$$

where q_b''' is related to q_b in Equation 2.31a as

$$\iiint q_b''' dV = q_b. \tag{2.37}$$

A newly freed electron will carry an excess energy of $h\nu - \mathcal{E}_b$, which will be deposited towards the energy of all free electrons with a rate of

$$q_{\text{pi}}''' = (h\nu - \mathcal{E}_b)\Gamma_{\text{pi}}. \tag{2.38}$$

Any absorption that do not result in ionization will instead excite bound electrons and serve as a heat source for the neutral particles with a rate of

$$q_{\gamma n}''' = (1 - \Phi)q_b''' = q_b''' - h\nu\Gamma_{\text{pi}}. \tag{2.39}$$

It is possible for a bound electron to simultaneously absorb multiple photons, a phenomenon known as multiphoton ionization [4]. This instead becomes a strongly nonlinear process, and the probability of multiple photons being absorbed by the same electron increases strongly with laser intensity. The minimum number of photons needed to completely remove an electron from its orbit is

$$N \equiv \left\lceil \frac{\mathcal{E}_b}{h\nu} \right\rceil. \tag{2.40}$$

The N-photon ionization rate is proportional to the N-th power of a particular measure of photon flux. In [5], that measure is the photon scalar flux, $\phi = \dfrac{I}{h\nu}$. However, since the laser intensity monotonic decreases with distance travelled, it is better to calculate the ionization rate using a path-averaged scalar flux:

$$\begin{aligned} \overline{\phi} &= \frac{\overline{I}}{h\nu} && (2.41\text{a}) \\ &= \frac{1}{h\nu s}\int_0^s I_0 e^{-\alpha s'} ds' && (2.41\text{b}) \\ &= \frac{I_0}{h\nu}\frac{1 - e^{-\alpha s}}{\alpha s} && (2.41\text{c}) \end{aligned}$$

The rate is then given by

$$\Gamma_{\text{pi}} = \min\left(S_N \overline{\phi}^N n, \frac{q_b'''}{Nh\nu}\right) \tag{2.42}$$

Here, S_N is the N-photon ionization rate coefficient and n is the neutral particle number density. The min function is to ensure that the actual ionization rate does not exceed the photon deposition rate. In order for Γ_{pi} to have the correct unit, the unit of S_N is $\text{m}^{2N}\ \text{s}^{N-1}$. If Equation 2.42 were to be in terms of another unit (for example, laser intensity) instead, then the unit of S_N must be changed accordingly. The term N-photon ionization cross section and the symbol σ_N are often used in literature to describe this coefficient [4] [5]; this nomenclature is not adopted here in order to ensure that all cross sections have units of area.

The rate of energy deposited into the neutrals and electrons are

$$\begin{aligned} q_{\text{pi}}''' &= (Nh\nu - \mathcal{E}_b)\Gamma_{\text{pi}} && (2.43\text{a}) \\ q_{\gamma n}''' &= q_b''' - Nh\nu\Gamma_{\text{pi}}. && (2.43\text{b}) \end{aligned}$$

For simplicity, bound electrons are assumed to only absorb N photons prior to removal from its parent nucleus. Above-threshold ionization, a phenomenon where an electron absorbs more than N photons, is also possible, but will be neglected here.

2.5 Summary

The system of equations have a total of 21 nonlinear variables, organized into four categories. The fluid equations for neutral particles are

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = -\frac{M}{N_A} \sum_r \Gamma_r \tag{2.44}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \rho \mathbf{u} \otimes \mathbf{u} = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g} + \sum_j^{e,i} \mathbf{f}_{jn} - \frac{M}{N_A} \sum_r \Gamma_r \mathbf{u}_r \tag{2.45}$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot \rho \mathbf{u} H = -\nabla \cdot \mathbf{q}'' + \nabla \cdot [\mathbf{T} \cdot \mathbf{u}] + \sum_j^{e,i,\gamma} q'''_{jn} - \frac{M}{N_A} \sum_r \Gamma_r E_r. \tag{2.46}$$

The fluid equations for electrons are

$$\frac{\partial n_e}{\partial t} + \nabla \cdot n_e \mathbf{u}_e = \sum_r \Gamma_r \tag{2.47}$$

$$\frac{\partial n_e \mathbf{u}_e}{\partial t} + \nabla \cdot n_e \mathbf{u}_e \otimes \mathbf{u}_e = \frac{-\nabla p_e - e n_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B})}{m_e} + \frac{1}{m_e} \sum_j^{i,n} \mathbf{f}_{je} \tag{2.48}$$

$$\frac{\partial n_e \mathcal{E}_e}{\partial t} + \nabla \cdot n_e \mathbf{u}_e \mathcal{H}_e = -e n_e \mathbf{u}_e \cdot \mathbf{E} + \sum_j^{i,n,\gamma} q'''_{je} + q'''_{\mathrm{pi}}. \tag{2.49}$$

The fluid equations for ions are

$$\frac{\partial n_i}{\partial t} + \nabla \cdot n_i \mathbf{u}_i = \sum_r \Gamma_r \tag{2.50}$$

$$\frac{\partial n_i \mathbf{u}_i}{\partial t} + \nabla \cdot n_i \mathbf{u}_i \otimes \mathbf{u}_i = \frac{-\nabla p_i + e n_i (\mathbf{E} + \mathbf{u}_i \times \mathbf{B})}{m_i} + \frac{1}{m_i} \sum_j^{e,n} \mathbf{f}_{ji} + \sum_r \Gamma_r \mathbf{u}_r \tag{2.51}$$

$$\frac{\partial n_i \mathcal{E}_i}{\partial t} + \nabla \cdot n_i \mathbf{u}_i \mathcal{H}_i = e n_i \mathbf{u}_i \cdot \mathbf{E} + \sum_j^{e,n} q'''_{ji} + \sum_r \Gamma_r \mathcal{E}_r. \tag{2.52}$$

Finally, Maxwell's Equations are needed to model the electric and magnetic fields.

$$\varepsilon_r \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\mu_0} \nabla \times \mathbf{B} - e(n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.53}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \tag{2.54}$$

Chapter 3

Numerical methods

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